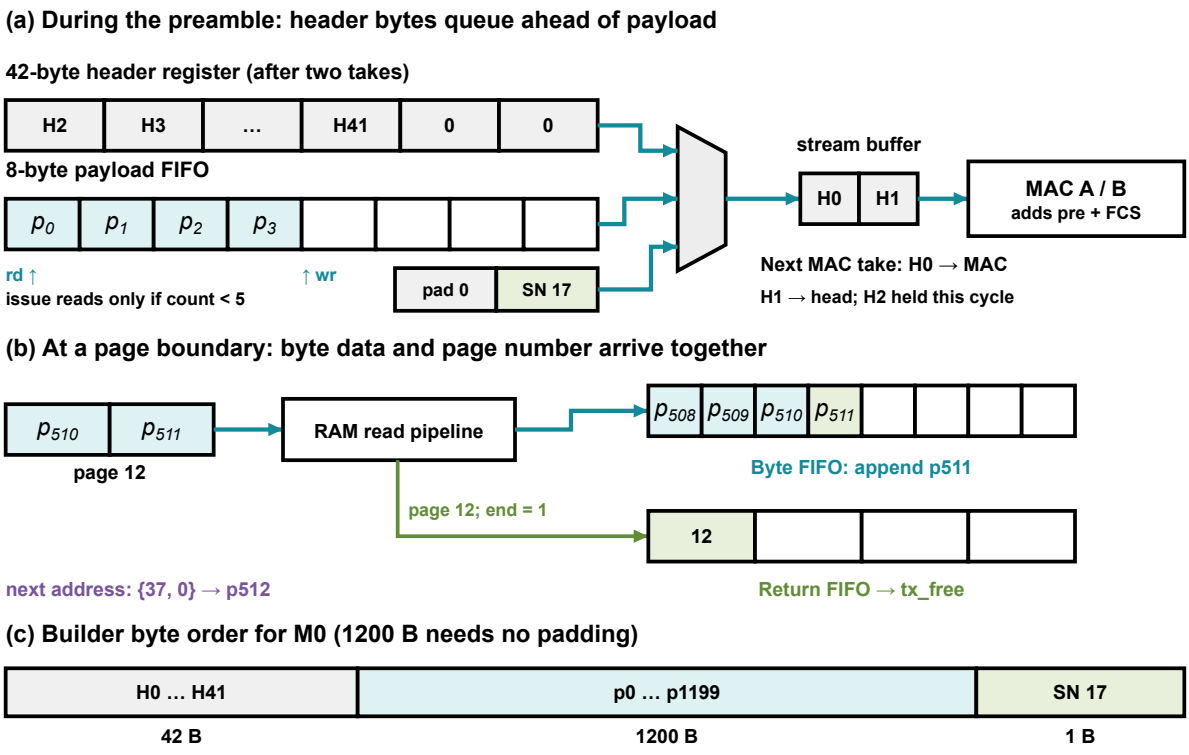
(a) During the preamble: header bytes queue ahead of payload
42-byte header register (after two takes)
H2
H3
…
H41
0
0
8-byte payload FIFO
p0
p1
p2
p3
rd ↑
issue reads only if count < 5
↑ wr
pad 0
SN 17
stream buffer
H0
H1
MAC A / B
adds pre + FCS
Next MAC take: H0 → MAC
H1 → head; H2 held this cycle
(b) At a page boundary: byte data and page number arrive together
p510
p511
page 12
RAM read pipeline
p508
p509
p510
p511
Byte FIFO: append p511
page 12; end = 1
12
next address: {37, 0} → p512
Return FIFO → tx_free
(c) Builder byte order for M0 (1200 B needs no padding)
H0 … H41
p0 … p1199
SN 17
42 B
1200 B
1 B